STA524A

N-channel
General purpose

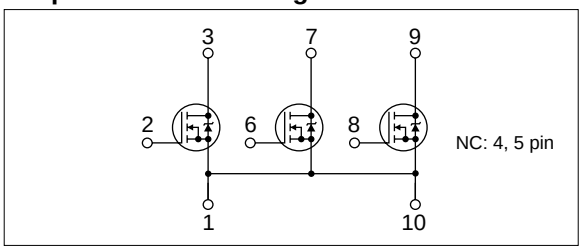
External Dimensions Ⓒ…STA (10-pin)

Absolute maximum ratings (Ta=25°C)

Symbol	Ratings	Unit
V_{DSS}	200	V
V_{GSS}	±20	V
I_D	±7	A
$I_{D(pulse)}$	±15 (PW≤100μs, duty≤1%)	A
E_{AS}*	55	mJ
I_{AS}	7	A
P_T	3.5 (Ta=25°C with all circuits operating, without heatsink)	W
	15 (Ta=25°C with all circuits operating, with infinite heatsink)	W
Tch	150	°C
Tstg	–40 to +150	°C

* : V_{DD}=25V, L=2mH, I_L=7A, unclamped, R_G=50Ω, see Fig. E on page 15.

■Equivalent circuit diagram



Electrical characteristics (Ta=25°C)

Symbol	Specification			Unit	Conditions
	min	typ	max		
$V_{(BR)DSS}$	200			V	I_D=100μA, V_{GS}=0V
I_{GSS}			±100	nA	V_{GS}=±20V
I_{DSS}			100	μA	V_{DS}=200V, V_{GS}=0V
V_{TH}	1.0		2.0	V	V_{DS}=10V, I_D=250μA
$Re_{(yfs)}$	4.0	7.0		S	V_{DS}=10V, I_D=3.5A
$R_{DS(ON)}$		280	350	mΩ	I_D=3.5A, V_{GS}=10V
		320	480	mΩ	I_D=3.5A, V_{GS}=4V
Ciss		420		pF	V_{DS}=10V
Coss		210		pF	V_{GS}=0V
Crss		90		pF	f=1.0MHz
$t_{d(on)}$		16		ns	I_D=3.5A, V_{DD}≒100V,
t_r		60		ns	R_L=28.6Ω,
$t_{d(off)}$		100		ns	V_{GS}=5V, R_G=25Ω,
t_f		70		ns	see Fig. 3 on page 16.
V_{SD}		1.0	1.5	V	V_{SD}=7A, V_{GS}=0V

Characteristic curves

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Datasheets for electronic components.